

Module 1

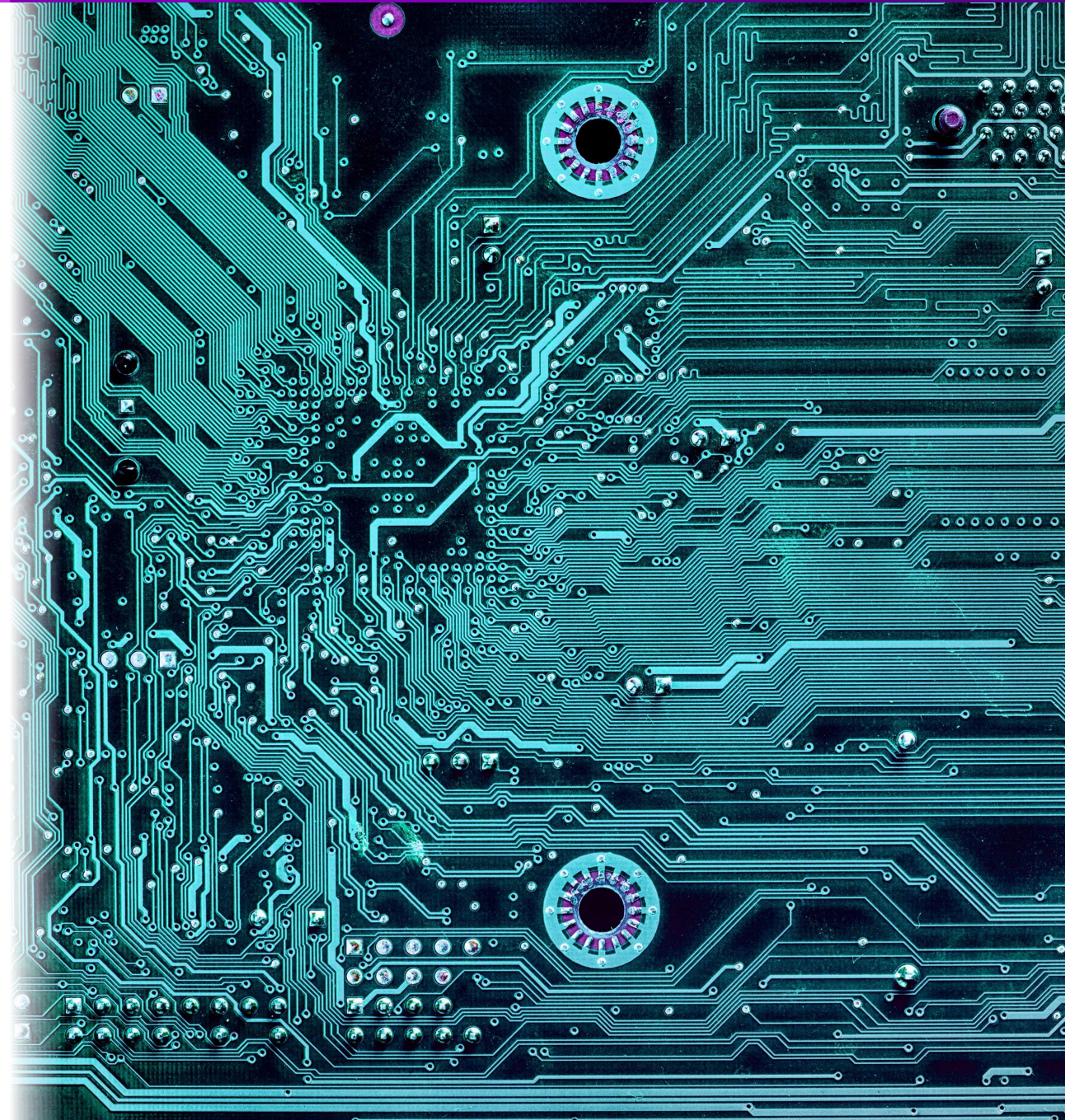
Signals and Interfaces

MTE 220 – Sensors and Instrumentation

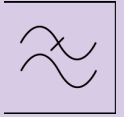


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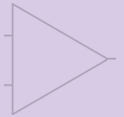
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Topics



1 – Signals and Interfaces



2 – New Devices and Circuits



3 – Sensors

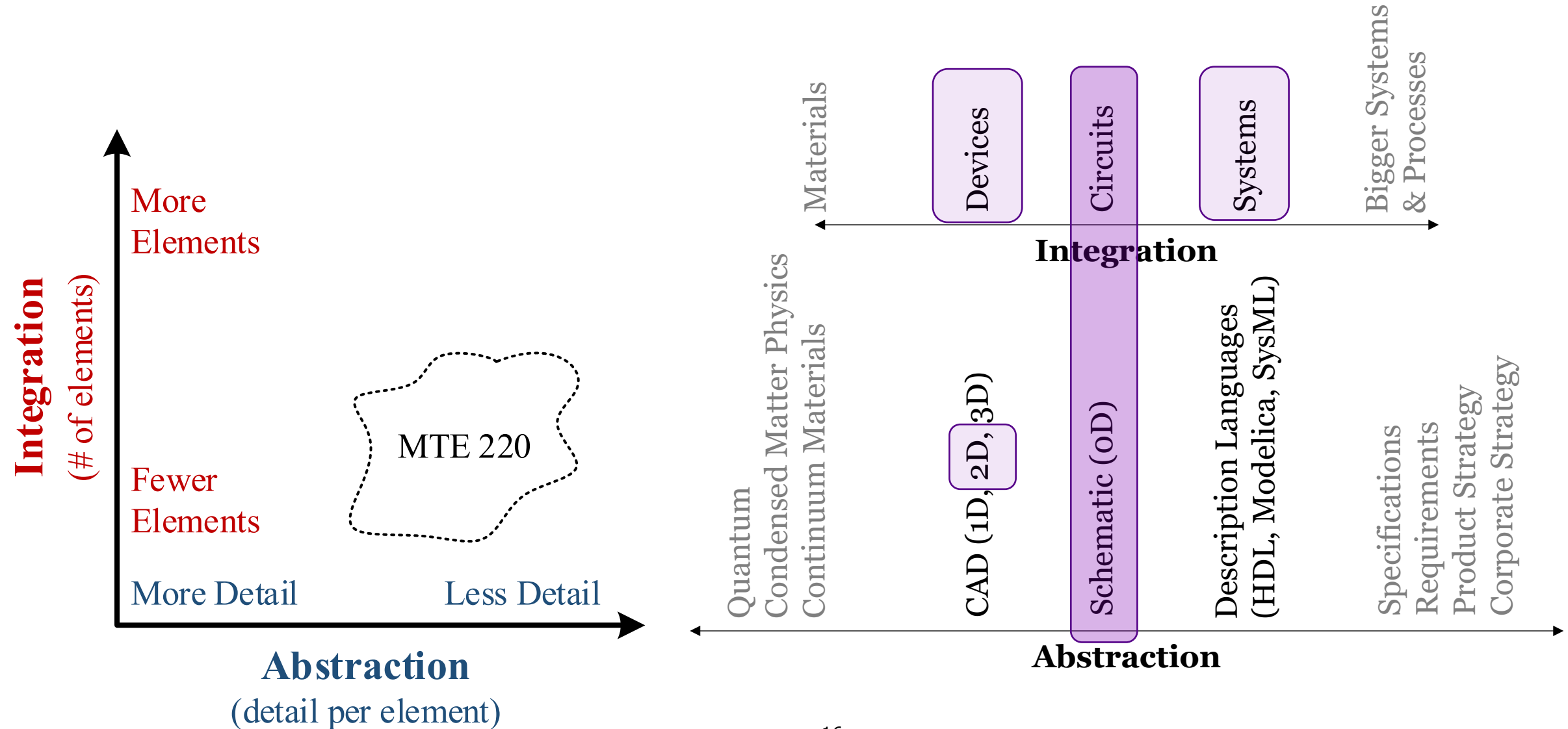


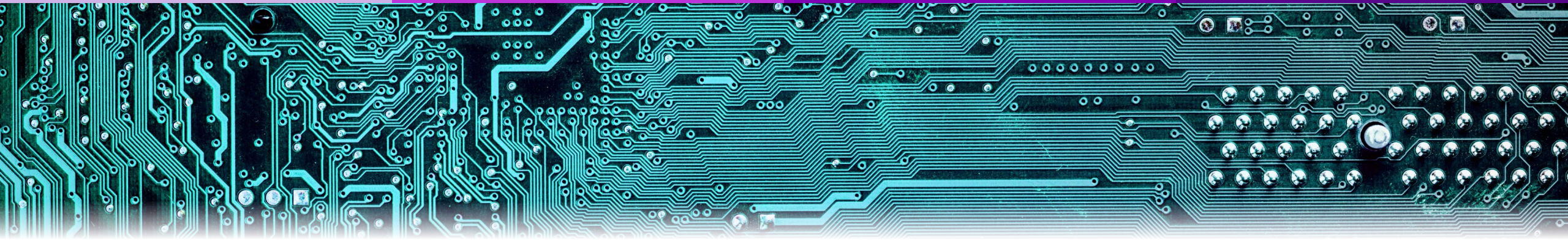
4 – Interfacing and Design

- 1.1 – Signal Basics
- 1.2 – Interfaces and Loading
- 1.3 – DC Operating Point and Signal Variation



Preamble: Integration and Abstraction

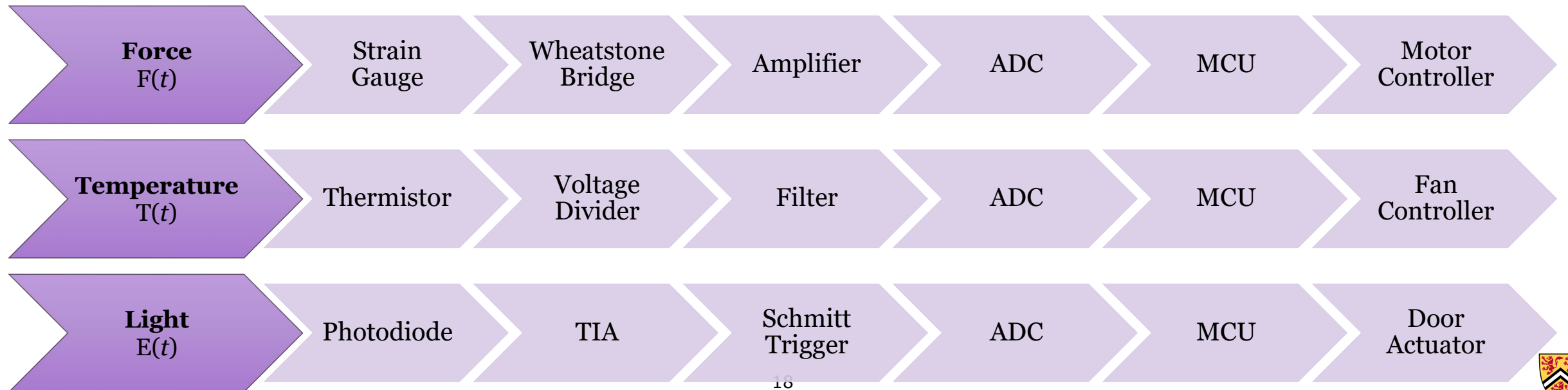




1.1 – Signal Basics

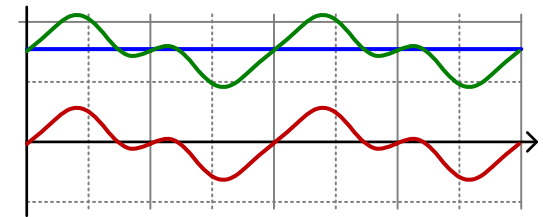
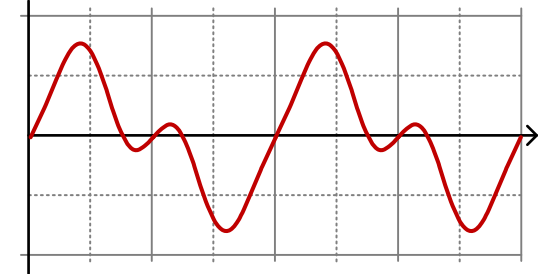
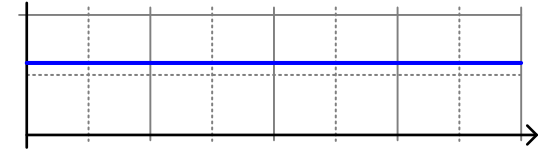
A Signal Is...

- A measurable variation that carries information
 - Variation may occur in time, space, frequency, or another state variable
 - Variation may be represented electrically, mechanically, optically, digitally, or otherwise
- In MTE 220: A voltage or current that represents information



AC and DC

- Direct Current (DC)
 - Original Meaning: Current and applied voltage do not change directions, but may fluctuate
 - Modern Meaning: Current or voltage is constant (also called “zero frequency component”)
- Alternating Current (AC)
 - Original Meaning: Current and applied voltage changes directions
 - Modern Meaning: Current or voltage is time-varying and has no constant offset (also called “non-zero frequency content”)
- We apply this terminology to any signal, regardless of physical meaning
 - E.g., “The **total signal** is the sum of the **DC offset** and **AC components**”

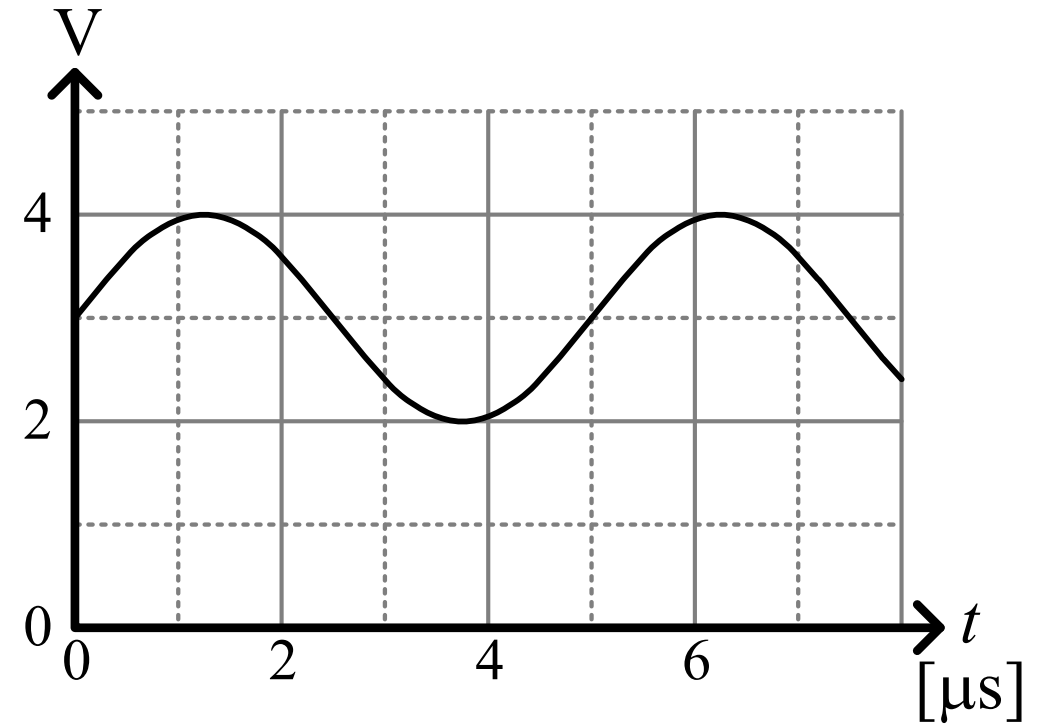


Waveform Nomenclature

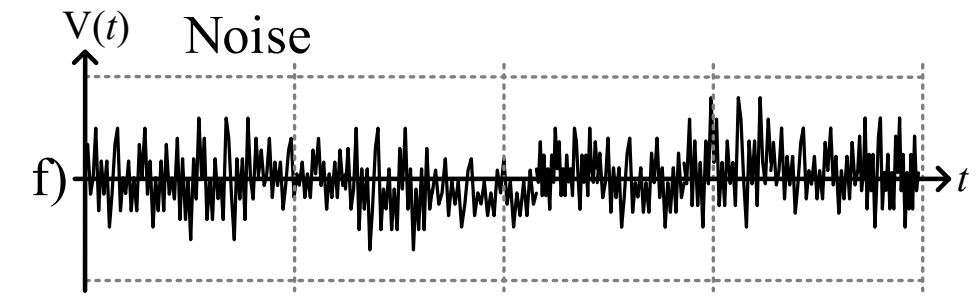
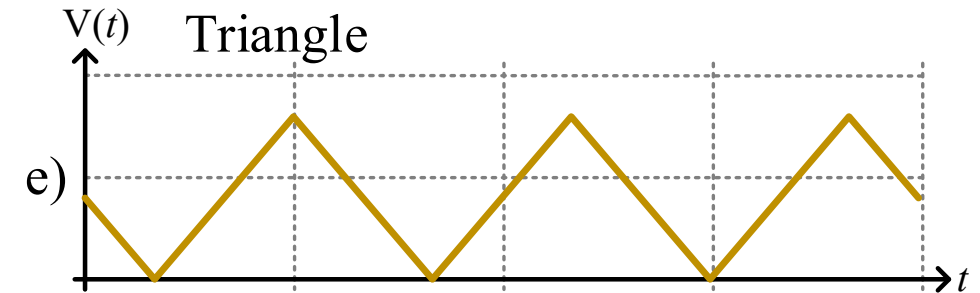
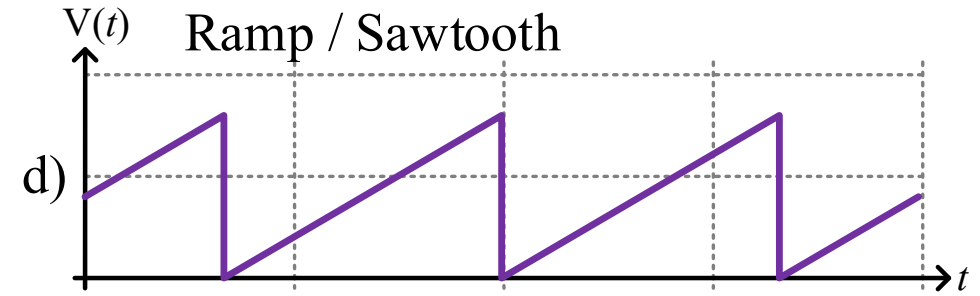
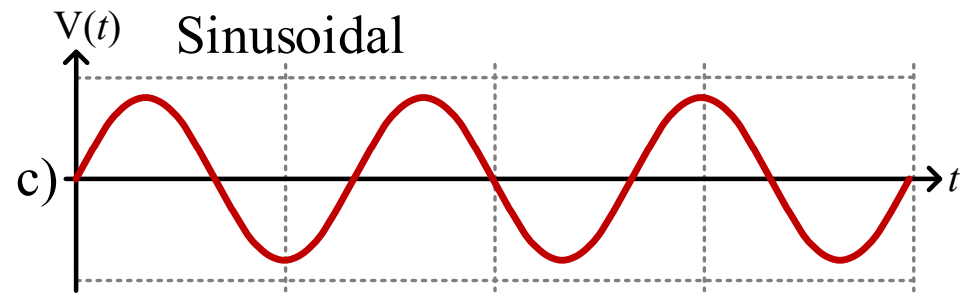
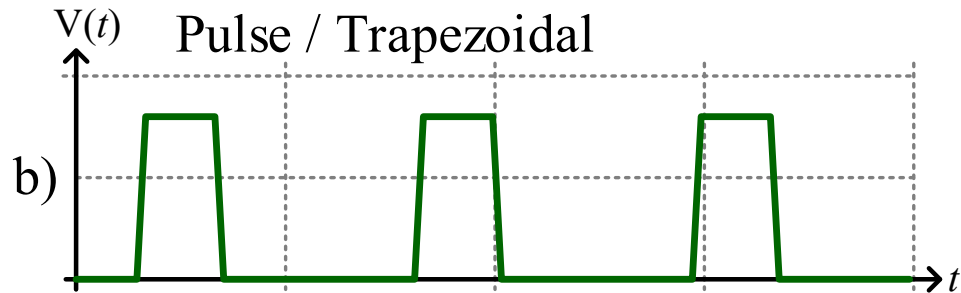
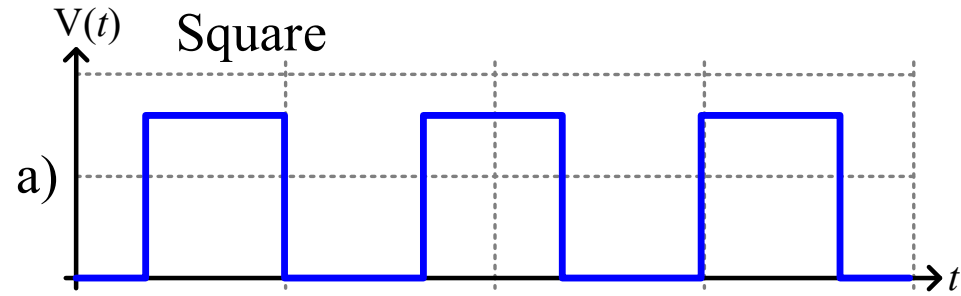
- From MTE 120:

Parameter	Value
V_{DC}	3 V
V_{pk}	1 V
V_{pk-pk}	2 V
T_o	5 μ s
f_o	200 kHz
ω_o	1.257 Mrad/s

- Analytically, $v(t) = 3 + \sin(2\pi \times 200k \times t)$
- Nomenclature:



Common Waveforms



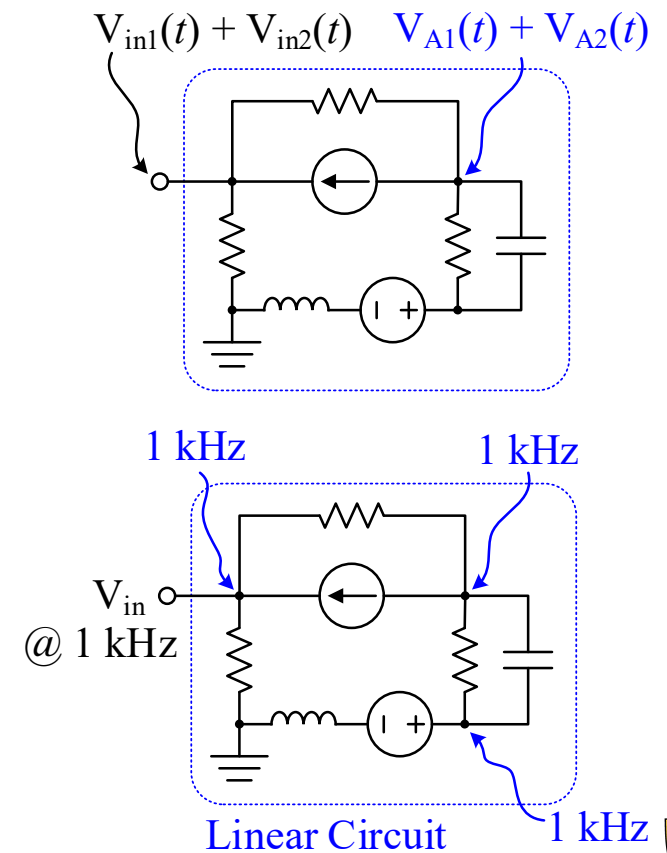
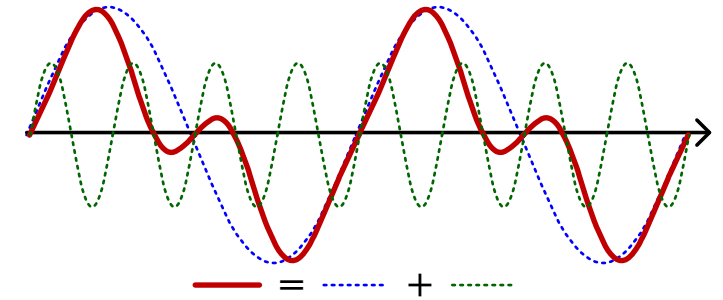
Linearity and Signals

We care so much about sinusoidal response because:

1. **Any signal is a sum of sine waves**
2. **Superposition applies to linear systems**
3. **Linear operations don't change the frequency**

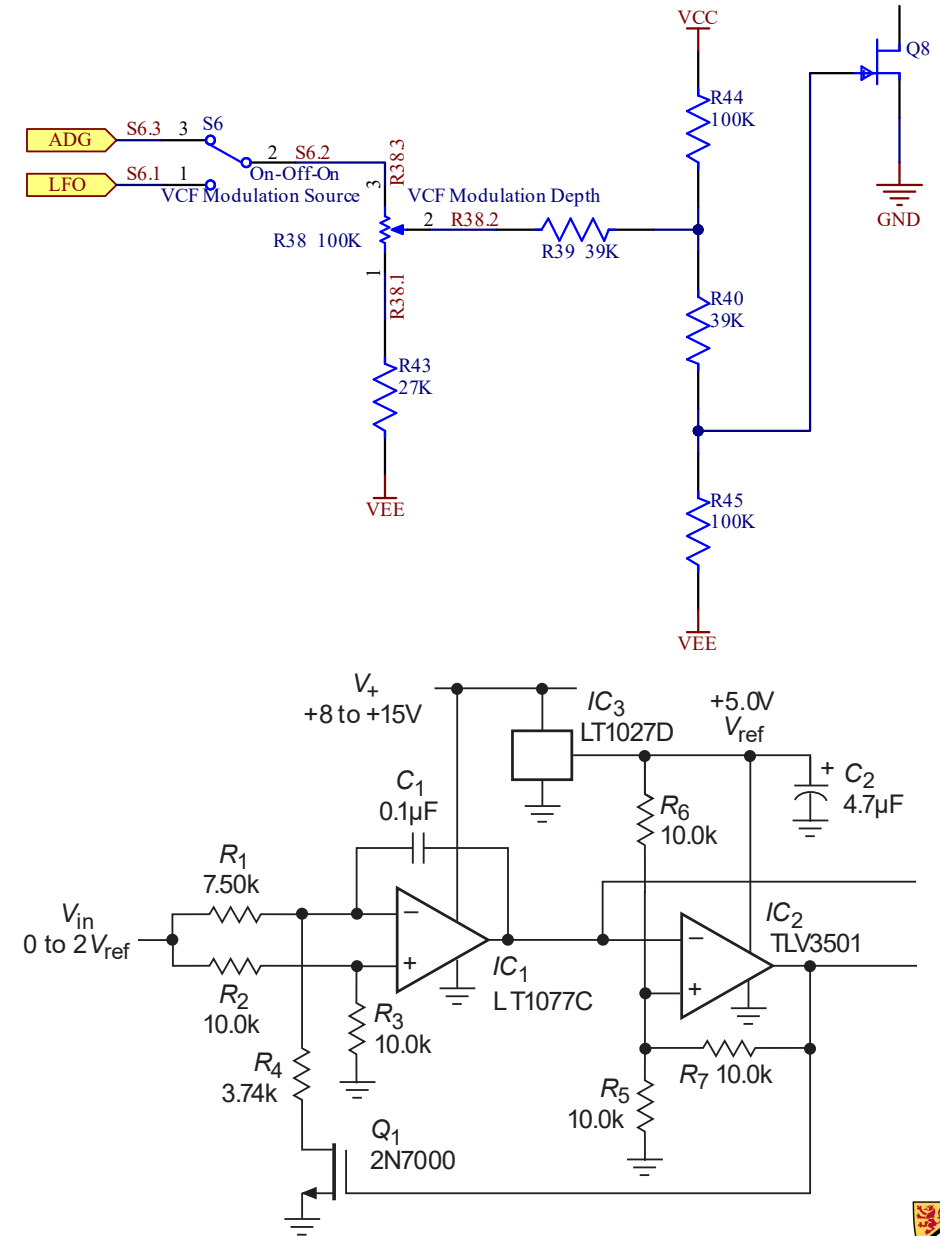
So, we tackle one frequency at a time in sinusoidal analysis

We can add the responses together to get the total response

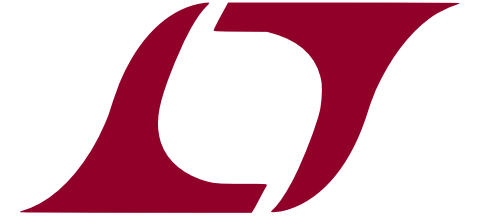


Voltage Rails

- Top Rail = max working voltage
 - V_{DD} or V_{CC} , or explicit: “+5 V”, “+3.3 V”
- Bottom Rail = min working voltage
 - V_{SS} , V_{EE} , or explicit: “−5 V”, “−9 V”
- Ground = 0 V; may also be the bottom rail
- Signals **must** remain between bottom and top rails (typically)
- Exceeding the rails can damage components



LTSpice Modeling Challenge



- Create voltage sources that output the waveforms on Slide 21
 - $1\mu = 1$ micro
 - $1\text{Meg} = 1$ million
 - $1\text{m} = 1\text{M} = 1$ milli
 - This button  lets you configure a “transient” (time-domain) simulation
- What are the challenges with square and ramp?
- What are the challenges with triangle?
- How did you make a noise source?
- Can you recreate the plot layout in Slide 21?

